



About DUCAT Institute – Pitampura

Introduction

DUCAT Institute is a well-known professional training organization providing industry-oriented IT education and career development programs. Established in the early 2000s, DUCAT has expanded across the Delhi–NCR region, with its Pitampura centre serving as a key hub for students and working professionals who want to upgrade their technical skills.

Vision & Mission

The institute's vision is to bridge the gap between academic knowledge and the real-world skills demanded by today's technology-driven industries. Its mission is to deliver high-quality, practical training that empowers learners to build rewarding careers in software development, web technologies, data analytics, digital marketing, and other emerging fields.

Courses and Specializations

DUCAT Pitampura offers a wide range of IT and professional courses, including:

- Full-Stack Web Development (HTML, CSS, JavaScript, React, Node.js)
- Programming in Java, Python, .NET, and C/C++
- ERP & SAP modules for business process management
- Data Science, Artificial Intelligence, and Cloud Computing
- Digital Marketing and e-commerce strategies

The curriculum is designed to combine classroom theory with live projects, case studies, and hands-on labs so that students gain practical, job-ready experience.

Key Strengths

- **Experienced Trainers:** Instructors with strong industry backgrounds ensure up-to-date, practical knowledge.

- **Industry-Relevant Curriculum:** Courses are continuously updated to match current technologies and market trends.
- **Placement Assistance:** Dedicated placement support helps students prepare résumés, practice interviews, and connect with hiring companies.
- **State-of-the-Art Facilities:** Modern computer labs and well-equipped classrooms provide a professional learning environment.

Location

The Pitampura centre is conveniently located near Kohat Enclave Metro Station in New Delhi, making it easily accessible to students across the city.

Commitment to Learners

DUCAT believes in lifelong learning and focuses on creating professionals who can adapt to fast-changing technologies. By emphasizing both technical excellence and practical skills, the institute has trained thousands of students and professionals, many of whom now work with leading national and international companies.

Acknowledgement

I would like to express my sincere gratitude to everyone who supported and guided me during my internship and the preparation of this report.

First and foremost, I thank **DUCAT Institute, Pitampura**, for providing me the opportunity to complete my internship in such a professional and inspiring environment. I am especially grateful to the institute's management and training faculty for sharing their knowledge and giving me practical exposure to real-time projects. Their valuable guidance helped me understand the current trends in web development and the importance of industry-oriented skills.

I wish to convey my heartfelt thanks to my **trainer Mr. Tarunendra Singh** whose constant encouragement, technical expertise, and constructive feedback were instrumental in the successful completion of this internship.

My sincere appreciation also goes to **Hindu College of Engineering** particularly the **Department of Computer Science** for supporting this training program and for their continuous motivation. I am thankful to my faculty coordinator for providing essential advice and timely suggestions throughout the project work.

I extend my gratitude to all the staff members, classmates, and friends who directly or indirectly contributed their time and assistance. Their cooperation and moral support played an important role in accomplishing my goals.

This internship has been a valuable experience that enhanced my professional skills and broadened my understanding of the industry.

-Khusbu chauhan

Skills learnt:

Technologies I used during my internship were as follows:

1. HTML
2. CSS
3. JavaScript
4. React.js
5. Next.js
6. Bootstrap
7. Tailwind
8. PostgreSQL Tools

I used during my internship were as follows:

9. Azure DevOps
10. Git/GitHub
11. Visual Studio Code

Architecture I used was the following:

12. Client-Server Architecture



HTML

HTML (Hyper Text Markup Language) is the standard language for structuring content on the web, serving as the backbone of user interfaces in React and Next.js projects. It was designed to be simple and efficient, enabling developers to create structured, semantic, and accessible web pages. In a React and Next.js project, HTML is dynamically generated and manipulated through React components and Next.js rendering methods, forming the client-side presentation layer in a client-server architecture. HTML is inherently flexible, defining the structure of elements like text, images, and forms, which are styled with CSS and enhanced with JavaScript (via React). It was developed by Tim Berners-Lee in the early 1990s and has evolved to HTML5, incorporating modern features like semantic tags, multimedia support, and APIs for dynamic content. In a React and Next.js project, HTML integrates seamlessly with React component-based architecture, where JSX (a syntax extension combining HTML-like tags with JavaScript) is used to define UI structures. Next.js enhances this by supporting server-side rendering (SSR) and static site generation (SSG), allowing HTML to be pre-rendered on the server for improved performance and SEO. This client-server synergy enables efficient communication between the client (browser rendering HTML) and server (Next.js API routes or external services). HTML's simplicity and English-like syntax make it accessible for beginners, while its flexibility supports a variety of use cases, from static landing pages to complex single-page applications. The ecosystem around HTML is vast, supported by tools like React libraries (e.g., Material-UI, Tailwind CSS for styling) and Next.js packages (e.g., NextAuth.js for authentication), which extend functionality for common development tasks.



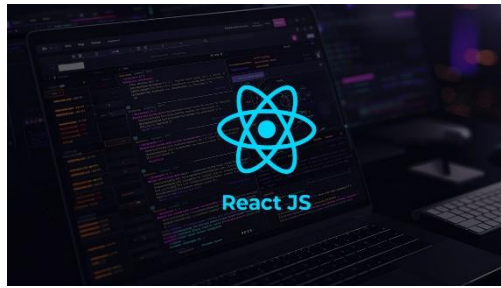
CSS

CSS (Cascading Style Sheets) is the standard language for styling and designing web content, playing a critical role in enhancing the visual presentation of React and Next.js projects. Designed for simplicity and flexibility, CSS allows developers to define the look and feel of HTML elements, ensuring an engaging and responsive user interface on the client side of a client-server architecture. CSS was developed by Håkon Wium Lie and Bert Bos in the mid-1990s and has evolved into CSS3, incorporating advanced features like animations, flexbox, grid layouts, and responsive design capabilities. In a React and Next.js project, CSS is used to style React components, often integrated through CSS modules, inline styles, or frameworks like Tailwind CSS, ensuring modular and maintainable styling. In the context of a React and Next.js project, CSS works seamlessly with React component-based architecture, where JSX combines HTML-like structures with JavaScript logic. CSS modules or styled-components are commonly used to scope styles locally to components, preventing style conflicts. Next.js enhances CSS integration by supporting server-side rendering (SSR) and static site generation (SSG), ensuring styled HTML is delivered efficiently to the client for optimal performance and SEO. In the client-server model, the client (browser rendering styled HTML) communicates with the server (e.g., Next.js API routes), creating a cohesive user experience. CSS is intuitive, with a straightforward syntax that makes it accessible for beginners, yet powerful enough to support complex designs through features like media queries and CSS variables. Its flexibility enables developers to create responsive, visually appealing interfaces adaptable to various devices and screen sizes. The ecosystem around CSS is robust, supported by tools like Tailwind CSS, Bootstrap, or libraries like Emotion, which provide pre-built solutions for styling challenges.



JAVA SCRIPT

JavaScript is a versatile, high-level programming language that powers the interactivity and logic of web applications in React and Next.js projects. Designed for flexibility and efficiency, JavaScript enables dynamic client-side functionality, making it a cornerstone of the client-server architecture. In a React and Next.js project, JavaScript drives the behaviour of user interfaces and handles server-side logic, ensuring seamless interaction between the client and server. JavaScript, created by Brendan Eich in 1995, is an interpreted language with just-in-time compilation, supporting both object-oriented and functional programming paradigms. Influenced by languages like C, Java, and Scheme, it is dynamically typed, treating everything as an object or function, which enhances its adaptability. In React and Next.js projects, JavaScript is used within JSX to define component logic and manage state, while Next.js leverages it for server-side rendering (SSR), static site generation (SSG), and API routes. In the client-server model, JavaScript powers the client-side (React components rendering dynamic HTML in the browser) and server-side (Next.js API routes or server components handling data processing). React (v18.3.1) uses JavaScript to create reusable, interactive UI components, while Next.js (v15.3.2), running in WSL (Windows Subsystem for Linux), optimizes JavaScript execution for performance and SEO through SSR and SSG. This ensures efficient communication between the client (browser) and server (Next.js backend). JavaScript's intuitive syntax, resembling English, makes it accessible for beginners, yet its robust features— like `async/await`, promises, and modern ES modules—support complex applications. Its vast ecosystem includes packages like `Lodash` for utility functions, `Axios` for HTTP requests, and `NextAuth.js` for authentication, extending functionality for common development tasks. The JavaScript community is vibrant and inclusive, with resources like MDN Web Docs, Stack Overflow, and meetups supporting developers at all skill levels.



REACT JS

React, developed by Facebook, is a widely used JavaScript library for building user interfaces. Known for its component-based architecture, React revolutionizes the way developers create interactive and dynamic web applications. With React, developers can break down their user interfaces into reusable components, enabling efficient development, code organization, and maintenance. React virtual DOM (Document Object Model) is a key feature that sets it apart. It uses a virtual representation of the actual DOM, allowing React to efficiently update and render components. By comparing the virtual DOM with the real DOM, React optimizes performance by minimizing unnecessary re-renders and updates. One of React core principles is unidirectional data flow, known as Flux or Redux architecture. This pattern provides a clear and predictable data flow, making it easier to manage application state and update the user interface accordingly. React unidirectional data flow simplifies debugging and helps maintain a consistent application state throughout the development process. Another strength of React is its strong ecosystem and vast community support. Numerous libraries, tools, and extensions are available to enhance React development, such as React Router for handling navigation, Redux for managing state, and Axios for making API requests. The extensive community support ensures that developers have access to a wealth of resources, tutorials, and solutions to address various challenges. React versatility is also worth mentioning. While primarily used for web development, React can be employed to build mobile applications using React Native. This flexibility makes React a popular choice for building cross-platform applications efficiently.



NEXT.JS

Next.js, built on top of React, is a full-stack JavaScript framework that enhances React capabilities for creating high-performance, scalable web applications. By leveraging React component-based architecture, Next.js enables developers to build reusable, modular UI components, streamlining development and improving code maintainability. With Next.js, React virtual DOM is seamlessly integrated, ensuring efficient updates by comparing the virtual representation of the UI with the actual DOM, minimizing re-renders, and optimizing performance for both server-rendered and client-side applications. Next.js enhances React unidirectional data flow, often paired with state management libraries like Redux , to ensure predictable data handling. This makes it easier to manage application state across server-side and client- side rendering, simplifying debugging and maintaining consistency in dynamic applications. For example, Next.js API routes allow developers to fetch data server-side, aligning with React data flow principles while reducing client-side overhead. The Next.js ecosystem amplifies React already robust community and tooling. Developers can use libraries like React Router (via Next.js built-in file-based routing), Redux for state management, and Axios or the native fetch API for data fetching within API routes or `getServerSideProps` . Next.js also supports React versatility by enabling developers to build not just web applications but also static sites, server-rendered apps, or even hybrid applications with incremental static regeneration. Additionally, React Native for Web can be paired with Next.js to create cross-platform experiences with shared codebases. Key Next.js features, like automatic code splitting, optimized image handling with the `next/image` component, and built-in SEO support, complement React strengths, making it easier to build fast, accessible, and search- engine-friendly applications. The vibrant React and Next.js communities provide extensive resources, tutorials, and packages to address development challenges.



BOOTSTRAP

Bootstrap is a popular open source front-end framework that simplifies the process of building mobile-first, responsive websites and web applications. Developed by Twitter, Bootstrap offers a collection of CSS and JavaScript components as well as pre-designed templates and styles that make it easy for developers to create visually appealing and functional interfaces. Responsive grid system is one of the key features of Bootstrap. It allows to adapt to different screen sizes and developers to create flexible layouts that automatically adjust and ensuring optimal viewing and usability across a wide range of devices. The grid system uses a 12- column layout, making it simple to organize content and establish responsive design patterns. Bootstrap also offers a rich set of CSS components and utilities that can be easily incorporated into web projects. These components include navigation bars, buttons, forms, alerts, modals, carousels, and more. With ready-to-use styles and classes, developers can quickly add and customize these components to match their design requirements. In addition to the CSS components, Bootstrap provides JavaScript plugins that enhance the functionality of the website or application. These plugins include features like dropdown menus, tooltips, modals, carousel sliders, and more. The JavaScript plugins seamlessly integrate with the CSS components, enabling interactive and dynamic user experiences.



TAILWIND

Next.js, built on React, is a powerful framework that enhances React component-based architecture to create high-performance, scalable web applications. By integrating Tailwind CSS, a utility-first CSS framework, developers can style React components with concise, reusable utility classes, eliminating the need for custom CSS files while maintaining flexibility and maintainability. React virtual DOM, seamlessly integrated into Next.js, ensures efficient UI updates by comparing the virtual representation with the actual DOM, minimizing re-renders and optimizing performance for both server-side and client-side rendering. Next.js complements React unidirectional data flow, often paired with state management libraries like Redux to ensure predictable state management across server and client rendering. Tailwind CSS enhances this by providing a consistent styling approach, allowing developers to apply styles directly in JSX using utility classes, which aligns with React declarative nature and simplifies debugging. For example, Next.js API routes enable server-side data fetching, reducing client-side overhead, while Tailwind's classes ensure the UI reflects the data efficiently. The ecosystem of Next.js, React, and Tailwind CSS is robust, supported by a vibrant community. Developers can use Next.js file-based routing (replacing React Router), Redux for state management, and Axios or fetch for API calls within Next.js API routes or `getServerSideProps`. Tailwind CSS integrates seamlessly, offering pre-built utilities for responsive design, animations, and theming, reducing reliance on external CSS libraries. Next.js versatility extends React capabilities, enabling static site generation, server-side rendering, or hybrid applications, while Tailwind's utility classes ensure consistent styling across web and mobile applications (using React Native for Web). Key Next.js features like automatic code splitting, optimized image handling with `next/image`, and built-in SEO support, combined with Tailwind's rapid styling workflow, empower developers to build fast, accessible, and visually appealing applications.



PostgreSQL

PostgreSQL, commonly called as Postgres, is a powerful open source relational database management system (RDBMS) known for its power and capabilities. With origins dating back to the 1980s, PostgreSQL has evolved into a feature-rich and reliable database solution used by organizations of all sizes. It adheres to the ACID (Atomicity, Consistency, Isolation, Durability) principles, ensuring data integrity and transactional reliability. One of its standout features is its support for advanced SQL queries and a wide range of data types, including arrays, JSON, and spatial data. PostgreSQL also provides an extensive set of built-in functions and extensions, allowing developers to enhance and customize its functionality to suit their specific requirements. Its support for concurrent access and scalability make it a preferred choice for high-performance applications and complex data workloads. Furthermore, PostgreSQL has a vibrant and supportive community that actively contributes to its development and provides extensive documentation and resources. With its stability, versatility, and active community, PostgreSQL continues to be a trusted and widely adopted database system for both small-scale projects and large enterprise applications. PostgreSQL is renowned for its proven standards, reliability, data integrity, robustness, scalability, and dedication to the open source community behind the software. offers performance and innovative solutions. It's no wonder that PostgreSQL has become the open source relational database of choice for many individuals and organizations.



AZURE DEVEOPS

Azure DevOps is a comprehensive platform that enhances the development of React-based Next.js applications styled with Tailwind CSS by providing integrated tools for version control, CI/CD pipelines, and agile project management. When combined with React component-based architecture, Next.js server-side rendering, and Tailwind's utility-first styling, Azure DevOps empowers teams to plan, code, test, and deploy scalable web applications efficiently. Azure Repos provides Git-based version control, enabling teams to collaborate on Next.js projects with features like pull requests and branch policies. React components and Tailwind-styled JSX files can be versioned, reviewed, and merged seamlessly, ensuring code quality. Azure Pipelines automates the build, test, and deployment process, allowing developers to create CI/CD workflows for Next.js applications. For example, a pipeline can build a Next.js app, run tests, and deploy it to Azure Static Web Apps or another hosting platform, leveraging Tailwind's optimized CSS output for fast load times. Azure Boards supports agile planning with Kanban boards and work item tracking, integrating with GitHub or Azure Repos to link commits and pull requests to tasks, ensuring traceability throughout the development lifecycle. Azure DevOps's ecosystem enhances React strengths with tools like GitHub Advanced Security for Azure DevOps for secret scanning, code scanning, and dependency scanning, ensuring secure codebases for Next.js projects. The platform's integration with Microsoft Entra ID reduces reliance on personal access tokens (PATs), enhancing security. Recent updates, such as improved Azure Boards + GitHub integration, allow automatic linking of pull requests and commits to work items, streamlining workflows for teams using GitHub alongside Azure DevOps. Additionally, Managed DevOps Pools simplify agent pool management, offering scalability and security for CI/CD pipelines. By combining Azure DevOps's automation and collaboration tools with React modular components, Next.js performance optimizations, and Tailwind's rapid styling, developers can deliver modern, secure, and visually appealing web applications with improved productivity and traceability.



Git/GitHub

Git is a distributed version control system used widely track changes over time and to manage source code in software development. Developed by Linus Torvalds, Git offers a decentralized approach, allowing developers to work on the same project simultaneously and merge their changes seamlessly. It provides a reliable and efficient way to handle branching and merging, enabling teams to collaborate effectively and manage complex development workflows. Git operates locally, meaning that developers can work offline and have full access to the complete history of the project. It tracks changes by creating snapshots of the codebase, allowing for easy navigation and retrieval of specific versions or commits. GitHub, on the other hand, is a web-based hosting platform built around Git. It provides a centralized and cloud-based repository for projects, allowing developers to store, share, and collaborate on code with ease. GitHub adds several features on top of Git, including a web interface for

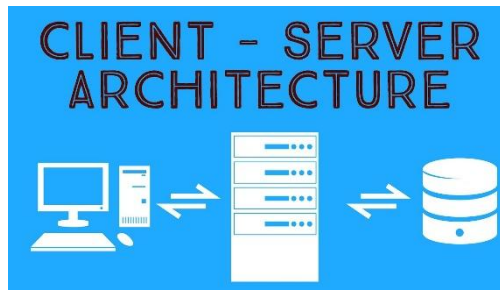
browsing repositories, pull requests for code review and collaboration, issue tracking, and project management tools. GitHub's social aspect fosters a vibrant community where developers can discover, contribute to, and learn from a vast collection of open-source projects. It also offers powerful integration options with various development tools, enabling continuous integration, automated testing, and deployment workflows. GitHub has become the go-to platform for both individual developers and organizations, providing a robust and scalable infrastructure for version control and collaboration. Together, Git and GitHub revolutionize the way developers work, promoting efficient collaboration, code sharing, and version control. Their combination offers a seamless and powerful ecosystem for managing source code, contributing to open-source projects, and facilitating team collaboration in the software development industry.



Visual Studio Code

Microsoft developed Visual Studio Code (VS Code) which is a powerful yet lightweight source code editor. It is highly extensible and supports a wide range of frameworks and programming languages. VS Code provides features such as syntax highlighting, code completion, debugging, version control integration, and an extensive library of extensions to enhance productivity and customization. Some key features of Visual Studio Code include:

- Cross-platform Support:** VS Code is available for macOS, Windows and Linux, allowing developers seamlessly across different operating systems to work.
- Debugging:** It offers a built-in debugger with support for various programming languages, allowing you to set breakpoints, step through code, inspect variables, and debug your applications effectively.
- Version Control Integration:** Git integration is built into VS Code, enabling you to manage your source code with features like commit, push, pull, diff viewing, and branch management.
- Extensions Ecosystem:** VS Code has a vast collection of extensions contributed by the community, which allows you to extend its functionality for specific languages, frameworks, tools, and workflows.
- Customization:** It offers extensive customization options, including themes, keyboard shortcuts, and settings, enabling you to tailor the editor to your preferences and optimize your development experience.
- IntelliSense:** Intelligent code suggestions and completion based on language semantics and your code context, making coding faster and more accurate is provided by VS Code.
- Integrated Terminal:** An integrated terminal allows to execute scripts, run commands and interact with your development environment without switching to an external terminal window.
- Intuitive User Interface:** It has a clean and intuitive user interface, with a sidebar for easy navigation, a panel for terminal output, and customizable layouts to suit individual preferences.



Client Server Architecture

Components of Client-Server Architecture

Client-side architecture refers to how the frontend of an application is structured and how its components interact within the browser environment. It plays a crucial role in creating dynamic, responsive, and scalable web applications. The architecture is typically broken down into three core components:

View – Interface to interact with users.

Controller – Manages user actions and updates the view and state accordingly.

Model – Manages application data and business logic on the client side.

View

The view is responsible for rendering the UI and presenting data to the user. It is the layer that directly interacts with users, capturing inputs like clicks, form submissions, or navigation actions. In client-side frameworks, views are usually components that dynamically update when the application state changes.

Controller

The controller acts as the logic layer between the view and the model. It handles user interactions triggered in the view and performs the necessary updates on the model or view accordingly. In client-side apps, the controller logic can be part of the component itself or handled through event handlers and lifecycle

Model

The model is responsible for managing the application's data, logic, and rules. It stores the state and updates it based on user actions or API responses. On the

client side, the model may include state objects, local storage, or front-end data caches.

Limitations

1. Limited Industry Experience: As an intern, you may have limited industry experience compared to experienced professionals.
2. Troubleshooting was complex
3. Scope of Assigned Tasks: Depending on the project and the specific internship program, I had limited control over the scope of the tasks assigned.
4. Integration Challenges: Integrating different components, systems, or APIs can present challenges, especially if there are compatibility issues, limited documentation, or third-party dependencies that may require additional troubleshooting and research.
5. Limited Exposure to Other Technologies: While focusing on React and Next on node.js, I had limited exposure to other programming languages, frameworks, or tools commonly used in the industry.
6. Time Constraints: As an intern, you may have limited time to complete projects and tasks.